D 10245 (**Pages : 3**) **Name**.....

Reg. No.....

FIFTH SEMESTER B.A./B.Sc. DEGREE EXAMINATION
NOVEMBER 2021

(CUCBCSS–UG)

Physics/Applied Physics

PHY 5B 08/APY 5B 09—PHYSICAL OPTICS AND MODERN OPTICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer in a word **or** *a phrase.*

Answer **all** *questions.*

Each question carries 1 *mark.*

1. The system matrix of a lens system is given by $\begin{bmatrix} 9/5 & 1/100 \\ 8 & 3/5 \end{bmatrix}$. The focal length of the combination is _____ units.
2. The inner part of optical fiber is called _____.
3. The construction of hologram made use of the principle of _____.
4. Two or more waves of the same frequency can maintain the same phase or constant phase difference over a distance and time are called _____.
5. The principal focal length of a zone plate is given by the equation _____.

True *or* False :

6. In N-slit Fraunhoffer diffraction pattern the principal maximum becomes sharper as N decreases.
7. The light ray incident at an angle of incidence greater than acceptance angle on a step index fiber will be refracted through cladding.
8. In order to observe good interference pattern, the distance of the screen from the sources must be large.
9. Colour appear on a thin film and on a soap bubble due to dispersion.
10. In a negative uniaxial crystal, velocity of O-ray is less than velocity of E-ray.

(10 × 1 = 10 marks)

Section B

Answer any **seven** *questions.*

Each question carries 2 *marks.*

11. Explain Fermat's principle of least time.
12. What are the advantages of using matrix method in paraxial optics?

Turn over

13. Describe the conditions to obtain sustained interference pattern.
14. What is the idea behind colour of thin films?
15. What is meant by overlapping spectra in the spectra of a grating?
16. Explain pulse dispersion in step index fiber.
17. Explain any two applications of holography.

(7 × 2 = 14 marks)

Section C

Answer any **five** *questions in paragraph of about* **half-a-page** *to* **one page**.

Each question carries 4 *marks.*

18. Obtain system matrix for thin film and hence obtain thin lens formula.
19. Describe Fresnel's biprism. Explain how the wavelength of light can be determined with the help of it.
20. With neat diagram, explain Michelson's interferometer. Describe how it can be used to determine the difference in wavelength of two waves.
21. Explain the principle and construction of a zone plate.
22. Draw the intensity distribution curve of Fresnel's diffraction at a straight edge and explain it.
23. Write brief note on (a) Quarter wave plate ; (b) Half wave plate.
24. Give the necessity of cladding in the optical fiber. Also mention any four applications of optical fiber.

(5 × 4 = 20 marks)

Section D

Answer any **four** *questions.*

Each question carries 4 *marks.*

25. Obtain a system matrix for a lens placed in air made of glass of refractive index 1.6 and radii of curvature 60 cm each. Find the focal length of the lens?
26. Two coherent sources are 0.2 mm apart and the fringes are observed on a screen 100 cm away. It is found that with a monochromatic source of light, the fourth bright fringe is situated at a distance of 10.8 mm from the central fringe. Calculate the wavelength of light.
27. Light of wavelength 5839 Å is reflected at near normal incidence from a soap bubble of refractive index 1.42. What is the least thickness of the film that will appear bright by reflection?
28. Newton's rings are formed with red light of $\lambda = 670$ nm. The radius of the 20th dark ring is found to be 1.1×10^{-2} m. Find the radius of curvature of the lens and the radius of 30th ring.

29. A plane transmission diffraction grating has 40,000 lines in all with grating element 12.5×10^{-5} cm. Calculate the maximum resolving power for which it can be used in the range of wavelength 5000 Å.

30. Find the thickness of (a) Quarter wave plate ; (b) Half wave plate, when light of wavelength 559 nm is used. Given $\mu_o = 1.55$, $\mu_E = 1.54$.

31. A step index fiber has a core diameter of 200 μm, its numerical aperture is 0.29. Calculate the number of propagating modes of an optical fiber with an operating wavelength of 859 nm.

(4 × 4 = 16 marks)

Section E

Answer in about **two pages**.

Answer any **two** *questions.*

Each question carries 10 *marks.*

32. Explain with necessary theory, the formation of Newton's rings in reflected light. How can we use this arrangement to determine the wavelength of light.

33. Give the construction and theory of plane transmission grating. Obtain the condition for absent spectra.

34. Explain the rectilinear propagation of light on the basis of Fresnel's half period zones.

35. What is the principle of holography ? Describe the recording and reconstruction process in holography with the help of suitable diagrams.

(2 × 20 = 20 marks)